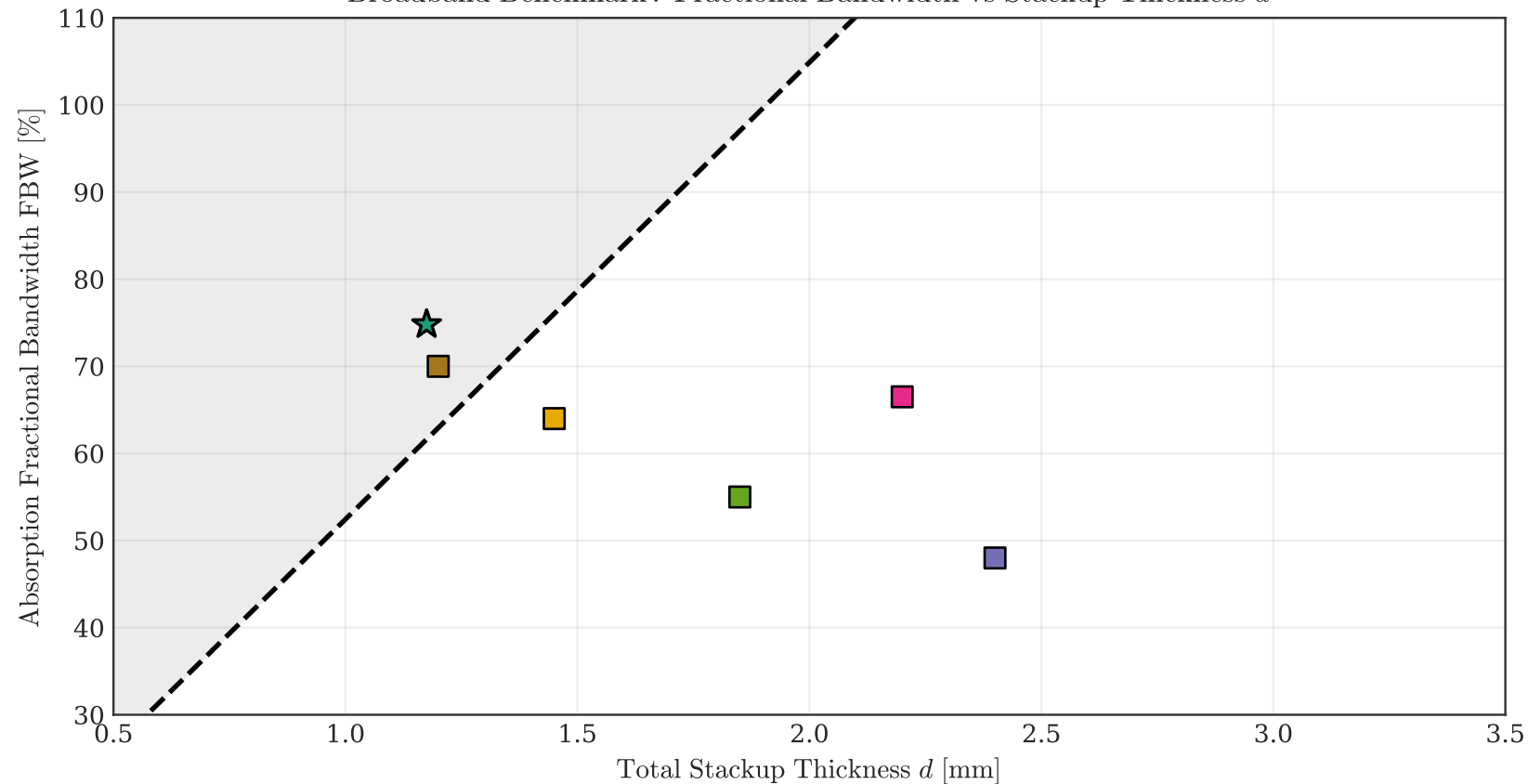


Broadband Benchmark : Fractional Bandwidth vs Stackup Thickness d



--- Theoretical Rozanov Causality Bound ($\rho_R = 1.0$)

Forbidden Physical Region ($\rho_R > 1.0$)

Smithetal. (2020)[IEEETAP]

Lietal. (2026)[Nano – MicroLett.]

Wangetal. (2024)[Adv. Mater.]

Maetal. (2025)[Adv. Funct. Mater.]

Zhangetal. (2025)[JCIS]

★ This Work : Golden Geometry #1 ($d = 1.175$ mm, $SE_T = 30.8$ dB, FBW = 74.8%)

★ This Work : Golden Geometry #2 ($d = 1.175$ mm, $SE_T = 30.8$ dB, FBW = 74.8%)